

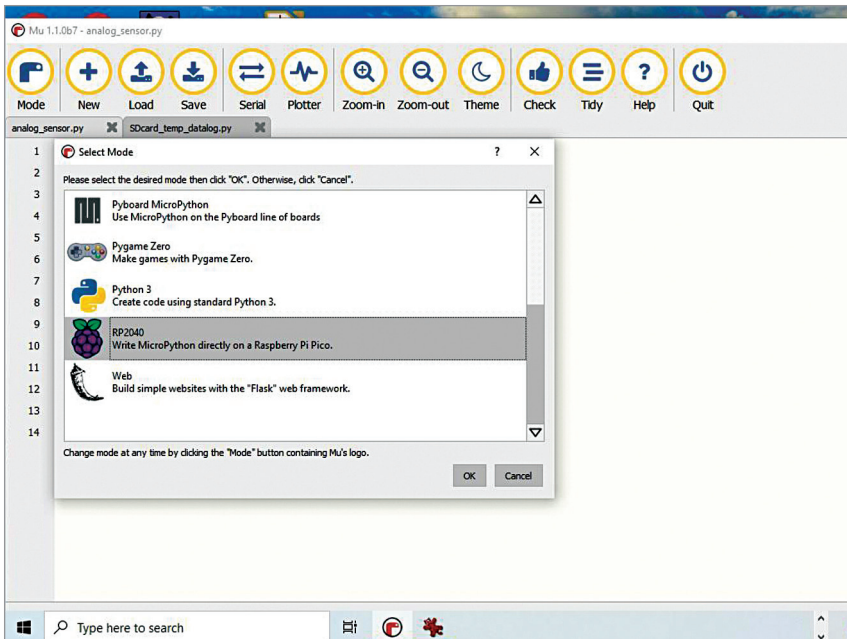


Fig. 6: Mode selection

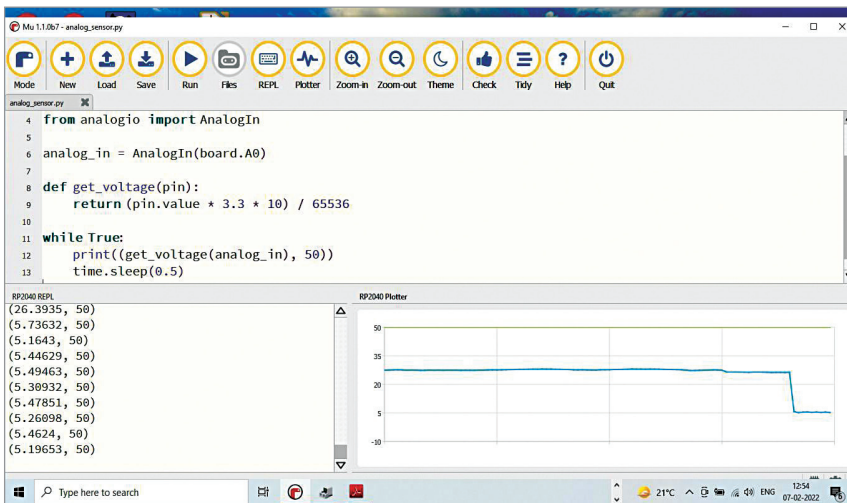


Fig. 7: LDR sensor output waveform

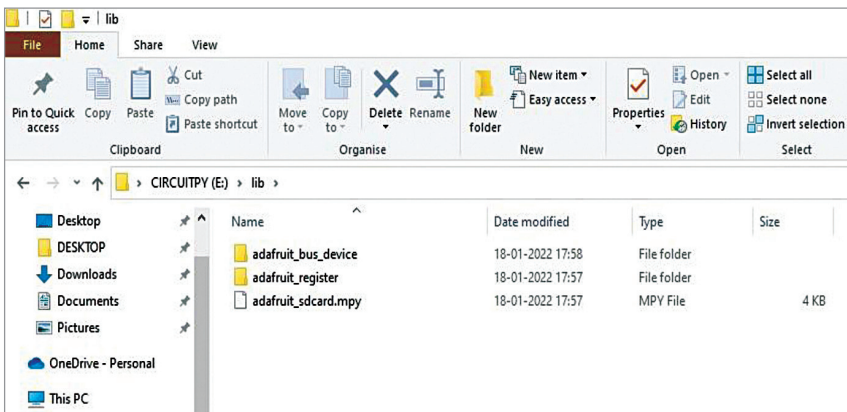


Fig. 8: CIRCUITPY drive

object and uncover it again, you will see the light intensity's arbitrary values vary between 0 and 32, with 0 being the darkest and 32 being the brightest light.

An additional feature of Mu is its ability to display the data waveform. When you select the Plotter option from the main menu, you will see the output waveform, as shown in Fig. 7. Similarly, you can replace LDR with any other analogue sensor like force sensor. The sensor data will be displayed on the screen along with the waveform.

Let us now explore the inbuilt temperature sensor on Raspberry Pi Pico board and store the temperature data on the microSD card that is normally used in your mobile phone. Here, we use CircuitPython libraries along with the main code (*Sdcard_temp_data.py*). You need to import adafruit libraries for SD card to interface it with the Raspberry Pi Pico board. For that download the libraries (Bundle for Version 7.x) from <https://circuitpython.org/libraries>.

Unzip the Bundle folder and copy *adafruit_bus_device* and *adafruit_register* folders, paste them into 'lib' folder in CIRCUITPY(E) drive. Also copy *adafruit_sdcard.mpy* file from the Bundle folder and paste into 'lib' folder, as shown in Fig. 8.

Open Mu editor and load the main code under *Code.py* file. Save the file and run it. If something goes wrong, you will get error message at the bottom of the Mu editor window. For example, if SD card is not connected properly to the circuit, you will get "OSError: no SD card" error message, as shown in Fig. 9.

If everything is fine, instead of error message you will see following text (see Fig. 10) below the Mu editor window: "The first line of text on SD card, Check temperature data on SD card"